

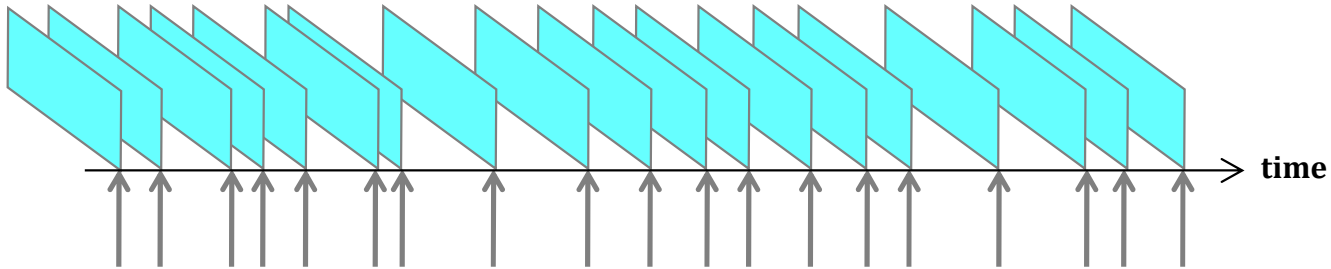
Workloads

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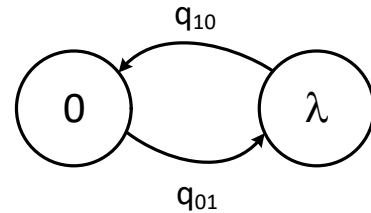
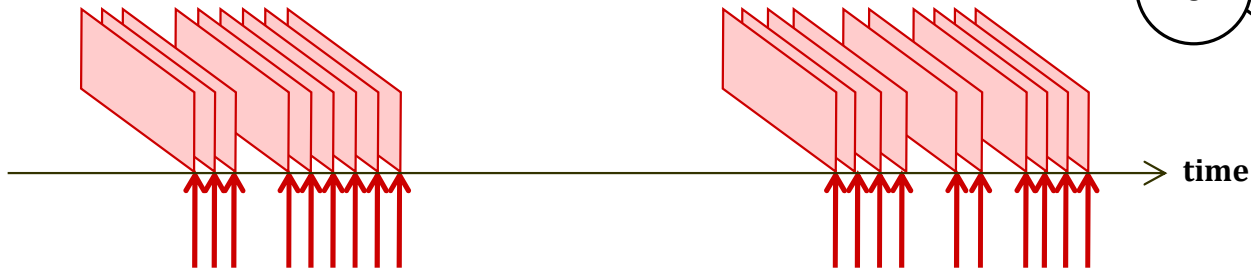
- The load placed into a system
- Two aspects
 - Arrival process: at what time it arrives
 - Job sizes: size of what arrives
- Arrival processes: Poisson, Interrupted Poisson, Markov Modulated Poisson Processes, ... - may have high **burstiness**
- Job sizes: exponential, Pareto, Bounded Pareto, ... - may have high **variability**
- In many cases, job sizes have high variability , sometimes following a **heavy-tailed** distribution

Modeling the traffic: packet arrivals and sizes

Poisson arrivals

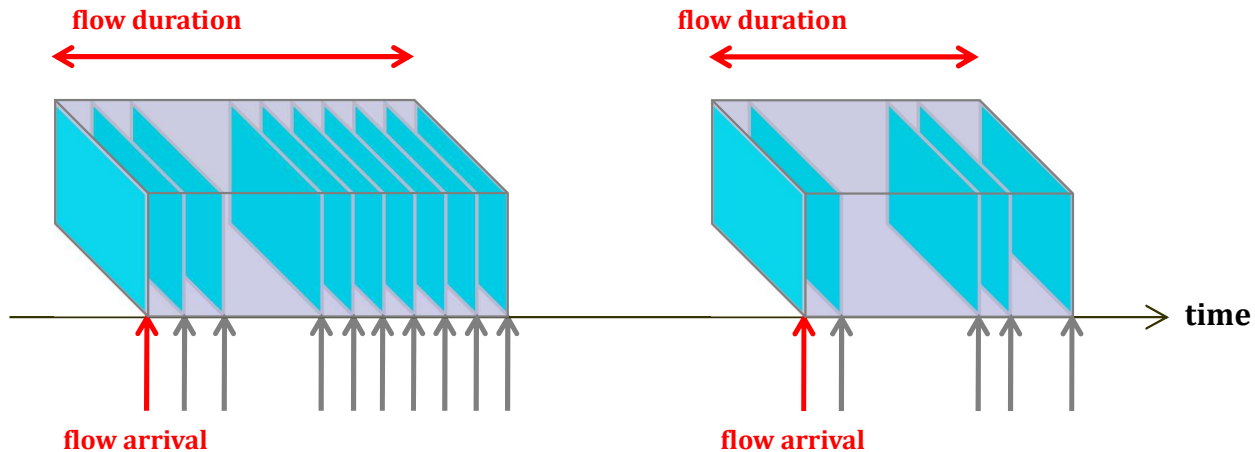


IPP arrivals



Modeling the traffic: flow and packet level

- Flow level: characterize flow arrivals and durations
- Packet level: characterize packet arrivals and sizes



Flow level examples: TCP sessions, VoIP calls

Pareto distribution

- Pareto distribution:

$$\bar{F}_X(x) = P\{X > x\} = x^{-\alpha}, \text{ for } x \geq 1$$

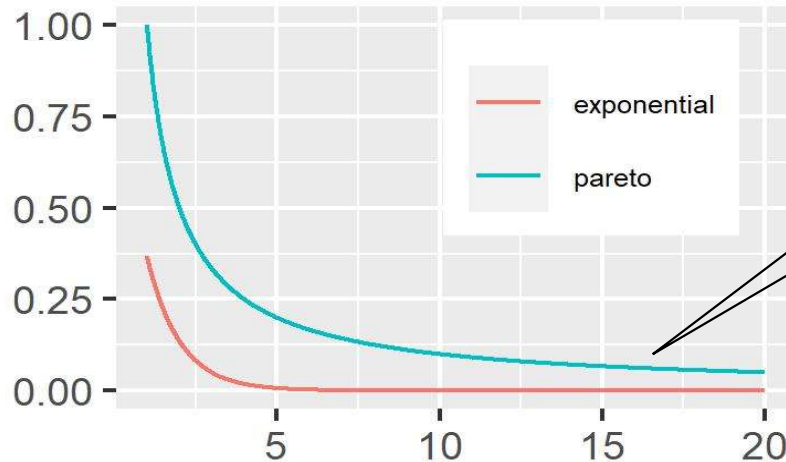
where $0 < \alpha < 2$.

$0 < \alpha \leq 1$:

$E(X)$
 $Var(X)$ } infinite

$1 < \alpha < 2$:

$E(X)$ finite
 $Var(X)$ infinite

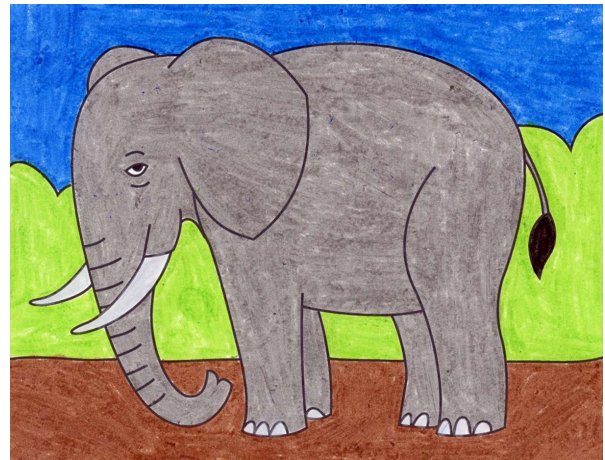
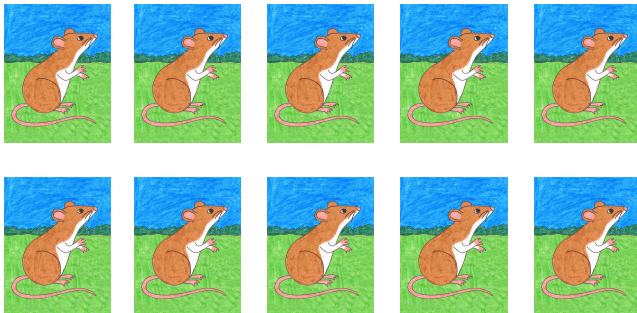


Exponential: $P\{X > x\} = e^{-\mu x}$
($\mu = 1$)

Pareto: $P\{X > x\} = x^{-\alpha}$
($\alpha = 1$)

Heavy-tail property

- A minuscule fraction of the very largest jobs comprise half of the total system load.
 - Pareto: when $\alpha = 1.1$, the largest 1% of the jobs comprise about half of the total load (exponential: largest 1% comprise about 5% of total)
 - Economics: the richest 1% of all people have more money between them than all the remaining of us combined
- **Elephants versus mice**: a few big elephants (big jobs) and many mice (little jobs)



Pareto job sizes

- Pareto job sizes are ubiquitous in computer networks and systems:
 - CPU lifetimes of UNIX jobs [Harchol-Balter, Downey 96]
 - Supercomputing job sizes [Schroeder, Harchol-Balter 00]
 - Web file sizes [Crovella, Bestavros 98], [Barford, Crovella 98]
 - IP flow durations [Shaikh, Rexford, Shin 99]
 - Wireless call durations [Blinn, Henderson, Kotz 05]
- Also ubiquitous in nature:
 - Forest fire damage – most fires cause little damage, largest few fires cause majority of damage
 - Earthquake damage – most earthquakes cause little damage, largest few earthquakes cause majority of damage
 - Human wealth – few (richest) people have most of the money

Bounded Pareto distribution

- Measured data has minimum and maximum job sizes. A bounded distribution is more appropriate to model measured data.
- Bounded Pareto distribution, $BP(L, U, \alpha)$:

$$f(x) = \alpha x^{-\alpha-1} \frac{L^\alpha}{1 - \left(\frac{L}{U}\right)^\alpha}$$

where $L \leq x \leq U$ and $0 < \alpha < 2$.

- All moments are finite.

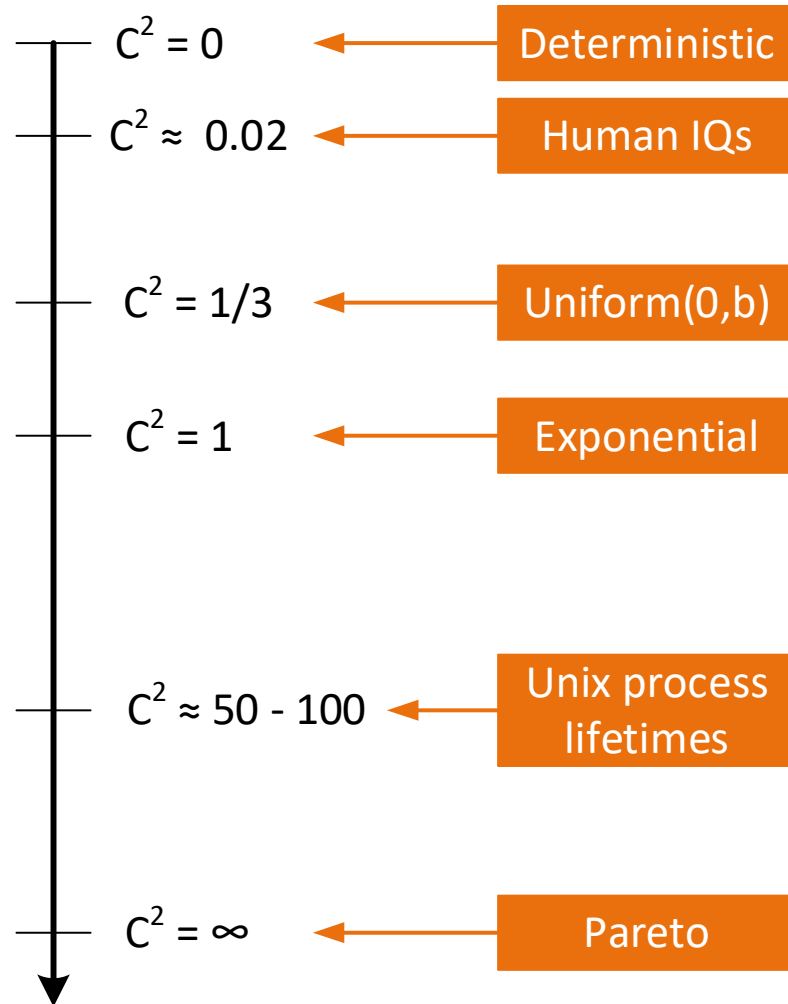
Variability in job sizes

- Squared coefficient of variation (SCV):

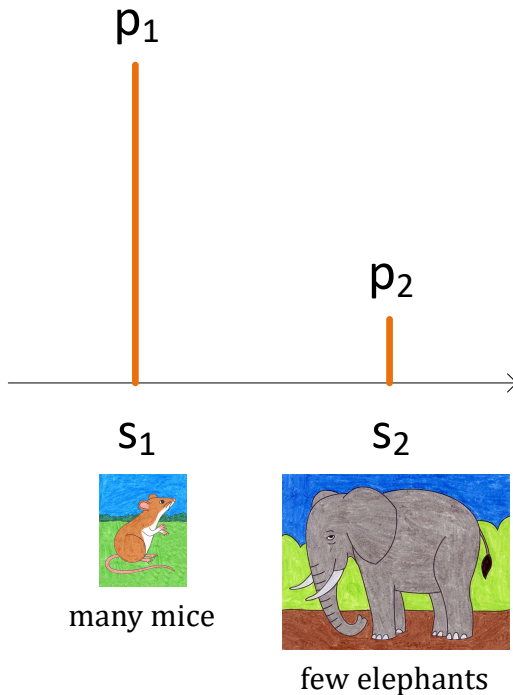
$$C_X^2 = \frac{\text{Var}[X]}{E[X]^2} = \frac{E[X^2]}{E[X]^2} - 1$$

- Can be seen as a normalized variance.

Variability in job sizes



Impact of variability on performance

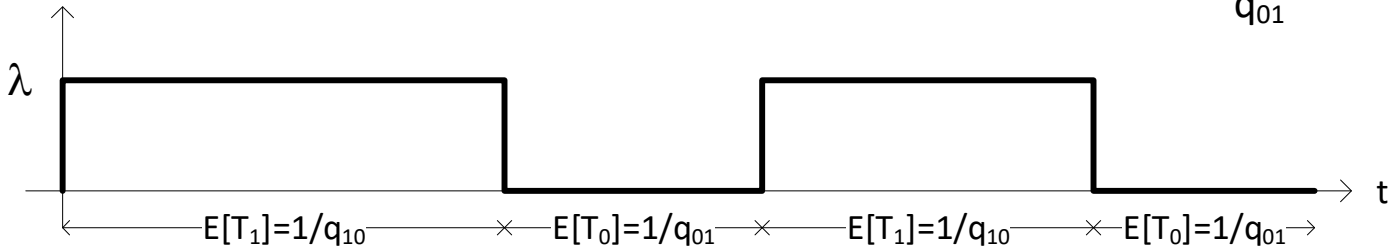
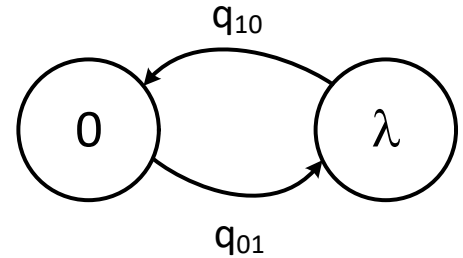


Queuing System ($\lambda/\mu = 0.5$)	C^2	W_q
M/M/1	1	2
M/Bimodal-1/1	2.25	3.25
M/Bimodal-2/1	24.75	25.75

Bimodal-1: $s_1 = 1$ with $p_1 = 0.9$; $s_2 = 11$ with $p_1 = 0.1$

Bimodal-2: $s_1 = 1$ with $p_1 = 0.99$; $s_2 = 101$ with $p_1 = 0.01$

Traffic burstiness



Average number of arrivals in state 1: $R = \frac{\lambda}{q_{10}}$

IDC (Index of Dispersion for Counts): $I_\infty = 1 + \frac{2R}{\left(1 + \frac{q_{01}}{q_{10}}\right)^2}$

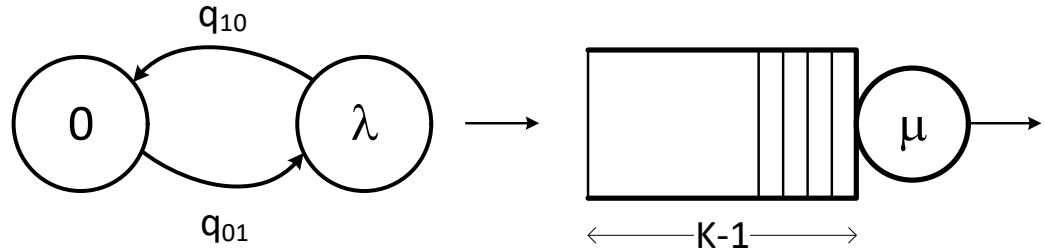
$I_\infty = 1$ for Poisson process

$\lambda \uparrow \Rightarrow I_\infty \uparrow$
 $R \uparrow \Rightarrow I_\infty \uparrow$
 $\frac{\pi_1}{\pi_0} \downarrow \Rightarrow I_\infty \uparrow$

$$\pi_0 = \frac{q_{10}}{q_{01} + q_{10}} \quad \pi_1 = \frac{q_{01}}{q_{01} + q_{10}} \quad \frac{\pi_1}{\pi_0} = \frac{q_{01}}{q_{10}}$$

Impact of burstiness on performance

IPP/M/1/K



$$R = \frac{\lambda}{q_{10}}$$

$$\begin{aligned} \lambda &= 10 \\ \pi_0 &= \pi_1 = 0.5 \\ K - 1 &= 10 \end{aligned}$$

R	IDC	loss
60	31	0.34
30	16	0.28
10	6	0.17
5	3.5	0.12
1	1.5	0.06
0.1	1.05	0.046

$$\rho = 0.9, \text{loss}(M/M/1/K) = 0.04372$$

R	IDC	loss
60	31	5.8×10^{-2}
30	16	4.4×10^{-2}
10	6	2.1×10^{-2}
5	3.5	9.6×10^{-3}
1	1.5	1.3×10^{-3}
0.1	1.05	3.0×10^{-4}

$$\rho = 0.5, \text{loss}(M/M/1/K) = 2.4 \times 10^{-4}$$